REVA UNIVERSITY • SCHOOL OF COMPUTING & INFORMATION TECHNOLOGY

AIML-401 • OFFICIAL FACULTY LECTURE MATERIAL

Course Material: Transformer Self-Attention & Multi-Head Projection

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1. Primary Teaching Points & Concept Excerpt

“Transformers discard sequential recurrence completely, replacing it with Scaled Dot-Product Self-Attention and Multi-Head Projections. This architecture achieves global receptive field at O(1) sequential step complexity, enabling massive parallelization during training across GPU clusters.”

2. Mathematical Formulation & Core Equations

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1. Scaled Dot-Product Attention:
    Attention(Q, K, V) = softmax( (Q • K^T) / sqrt(d_k) ) • V

2. Multi-Head Attention Projection:
    MultiHead(Q, K, V) = Concat(head_1, ..., head_h) • W^O
    where head_i = Attention(Q • W_i^Q, K • W_i^K, V • W_i^V)

3. Sinusoidal Positional Encoding:
    PE(pos, 2i)   = sin( pos / (10000^{2i / d_model}) )
    PE(pos, 2i+1) = cos( pos / (10000^{2i / d_model}) )
```

3. 4-Stage Algorithmic Architecture Pipeline

Stage	Pipeline Component	Mathematical / Architectural Operation	Key Objective
1	Q, K, V Projection	Multiply token embeddings by learnable matrices W^Q, W^K, W^V	Subspace alignment
2	Affinity Scoring	Matrix multiply Q • K^T scaled by 1/sqrt(d_k)	Similarity distribution
3	Softmax Weights	Apply row-wise softmax normalization	Attention probability matrix
4	Weighted Sum & Residual	Multiply attention probabilities by V, apply LayerNorm & residual	Contextual token vector

4. High-Yield Question & Marking Rubric for Midterm IA-2 & SEE Exams

Expected University Exam Question (8 Marks): *'Why is the scaling factor 1/sqrt(d_k) crucial in dot-product attention? Derive the multi-head attention mechanism and explain how positional encodings preserve sequential ordering in non-recurrent architectures.'*

Official Faculty Marking Rubric:

• 2 Marks: Mathematical proof that large d_k pushes softmax into extremely small gradient regions.

• 3 Marks: Mathematical formulation of Multi-Head Attention and projection matrices.

• 3 Marks: Linear relationship proof for sinusoidal positional encodings: PE(pos+k).